



Exponential Lower Bounds for the Pfaffian Number of Graphs¹²

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Indian Institute of Technology Indore

TIFR STCS Seminar

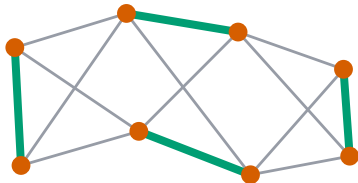
1 September 2026

¹Joint work with Priyanshu Pant

²Dedicated to the memory of Prof. R. B. Bapat

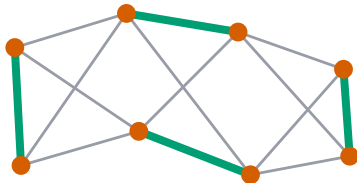
What is this talk about?

1. Counting perfect matchings in a graph



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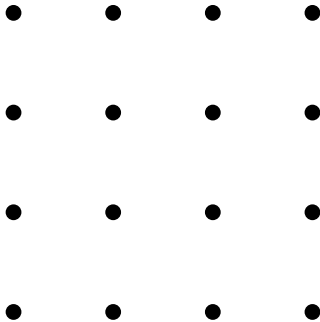


2. Permanent & Determinant

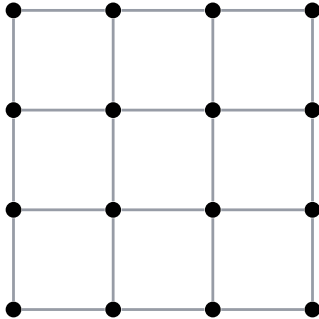
The non-identical twins



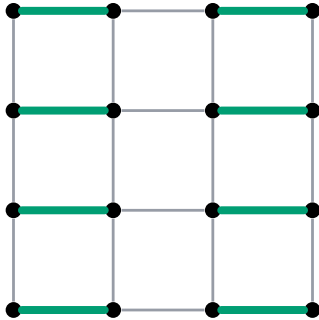
A graph has **vertices**



A graph has vertices and **edges**

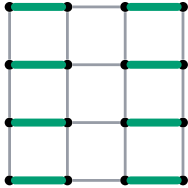


A perfect matching

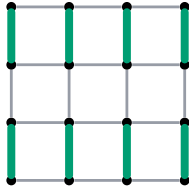


disjoint edges covering every vertex exactly once

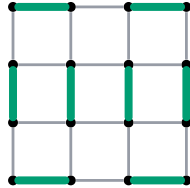
Graphs can have many **perfect matchings**



M_1

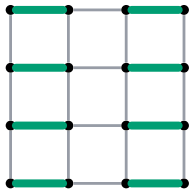


M_2

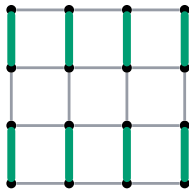


M_3

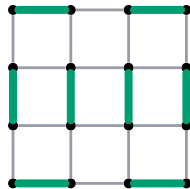
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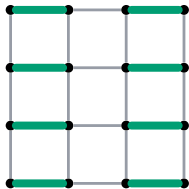
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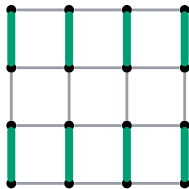
M_3

In fact, this graph has **36 perfect matchings**.

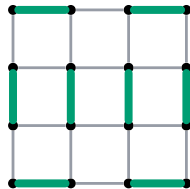
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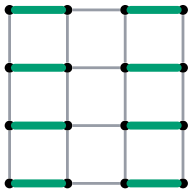
M_2



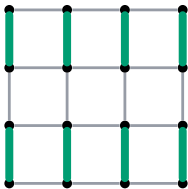
M_3

In fact, this graph has **36 perfect matchings**. But, can we count them efficiently?

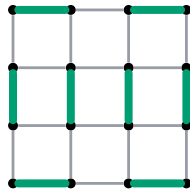
Graphs can have many **perfect matchings**



M_1



M_2



M_3

In fact, this graph has **36 perfect matchings**. But, can we count them
efficiently?

No. In general, this is hard.

But can we count efficiently
for some **nice classes of graphs**?

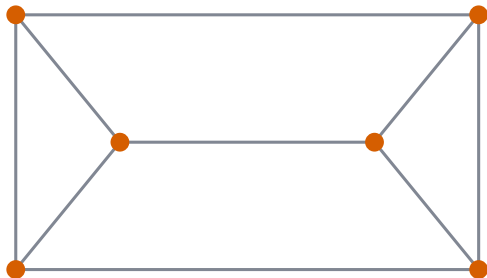
Yes.

In particular, for **all planar graphs**.

More generally, whenever G admits a **Pfaffian orientation**.

What is a Pfaffian orientation?

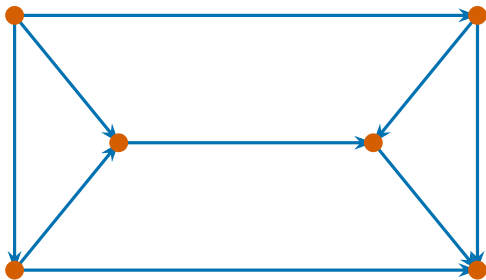
What is an orientation?



Given a graph G .

What is a Pfaffian orientation?

What is an orientation?

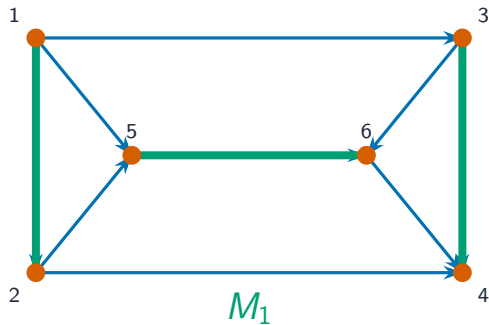


an orientation D

Given a graph G .
An **orientation** D assigns
a direction to every edge
of G .

What is a Pfaffian orientation?

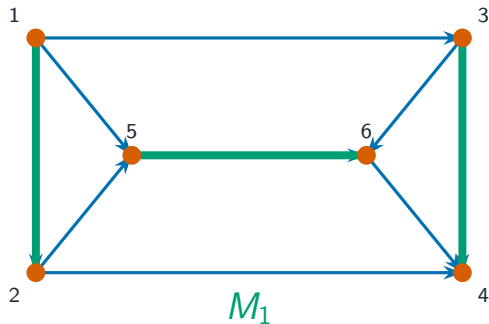
The sign of a perfect matching



$$M_1 = \{12, 34, 56\}.$$

What is a Pfaffian orientation?

The sign of a perfect matching



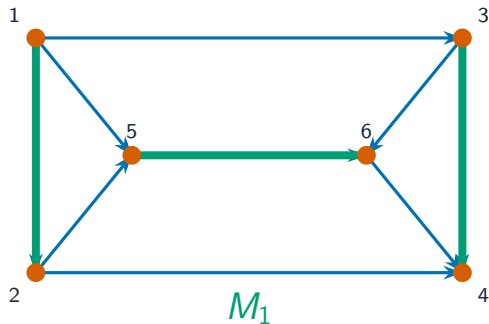
$$M_1 = \{12, 34, 56\}.$$

Reading tail then head we get a permutation

$$\pi_D(M_1) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix}.$$

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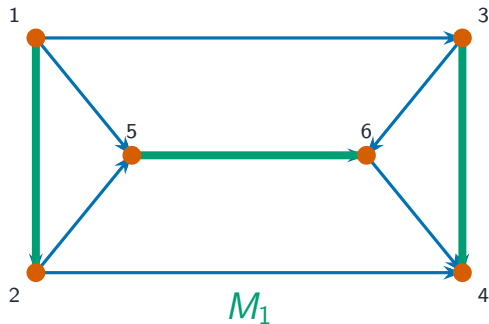
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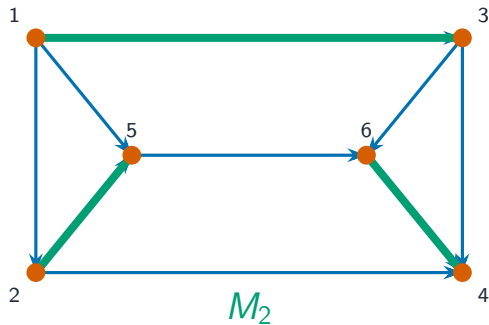
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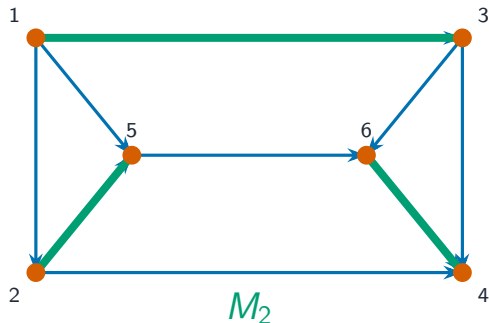


Similarly, consider

$$M_2 = \{13, 25, 46\}.$$

What is a Pfaffian orientation?

The sign of a perfect matching



Similarly, consider

$$M_2 = \{13, 25, 46\}.$$

Reading tail then head gives

$$\pi_D(M_2) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 2 & 5 & 6 & 4 \end{pmatrix}.$$

$$\text{sgn}_D(M_2) = -1.$$

What is a Pfaffian orientation?

The sign of a perfect matching

Formally, given some orientation D of a graph, and a perfect matching

$$M = \{u_1 v_1, u_2 v_2, \dots, u_m v_m\},$$

where in D each matching edge is oriented $u_i \longrightarrow v_i$.

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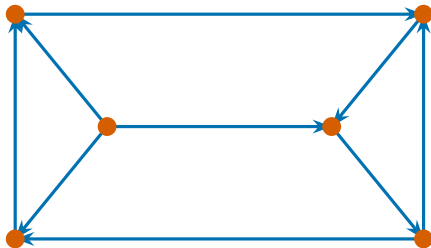
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and the sign of the perfect matching M with respect to D is

$$\text{sign}_D(M) = \text{sign}(\pi_D(M))$$

***Note** that the ordering of the edges of M does not matter.

What is a Pfaffian orientation?



a Pfaffian orientation D^*

Pfaffian orientation

An orientation in which all perfect matchings have the same sign.

$$\text{sgn}_D(M) = \varepsilon \quad \forall M \in \mathcal{M}(G), \\ \varepsilon \in \{+1, -1\}.$$

and, G is called Pfaffian.

Why Pfaffian orientations?

The Pfaffian

For a directed graph D , assign a variable x_{ij} to each edge ij .

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The Pfaffian

For a directed graph D , assign a variable x_{ij} to each edge ij .

Define its **weighted skew-adjacency matrix** A_D by

$$(A_D)_{ij} = \begin{cases} x_{ij}, & \text{if } i \rightarrow j, \\ -x_{ij}, & \text{if } j \rightarrow i, \\ 0, & \text{otherwise.} \end{cases}$$

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Set all edge variables to 1. Then

$$|\text{Pf}(A_D)| = \# \text{PM}(G).$$

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How is this efficient?

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For any even-order skew-symmetric matrix, Cayley's identity says

$$\text{Pf}(A_D)^2 = \det(A_D) \implies \text{Pf}(A_D) = \sqrt{\det(A_D)}.$$

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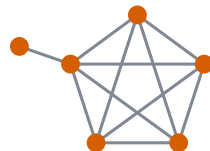
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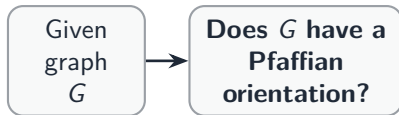
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But there are **non planar Pfaffian graph** as well.



K_5 with a pendant vertex

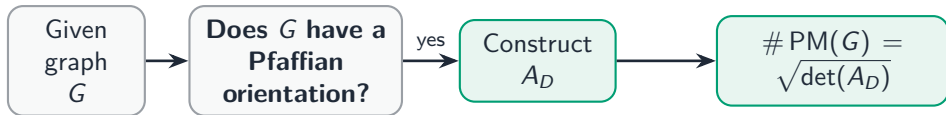
So far



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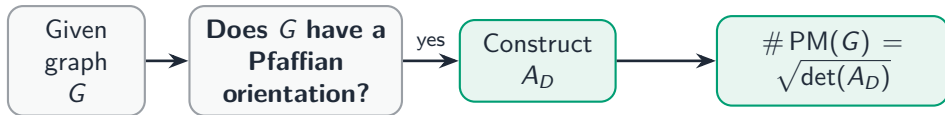


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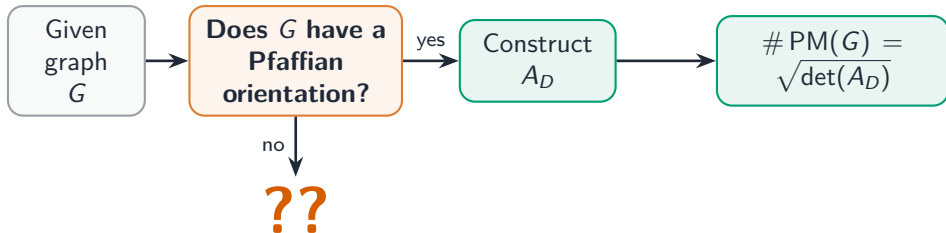
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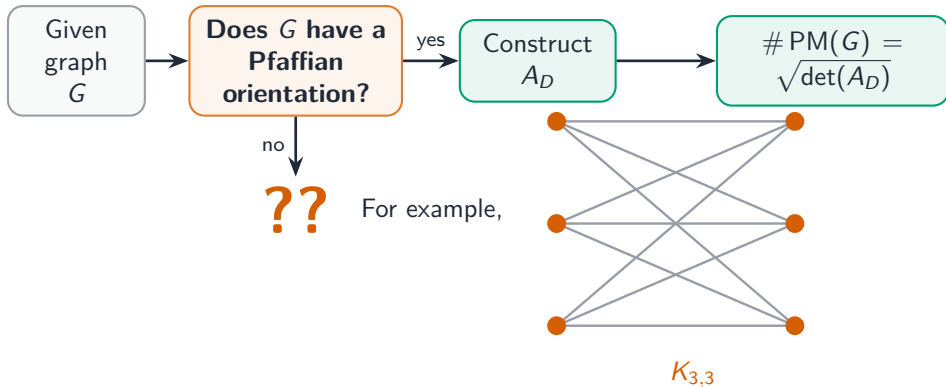
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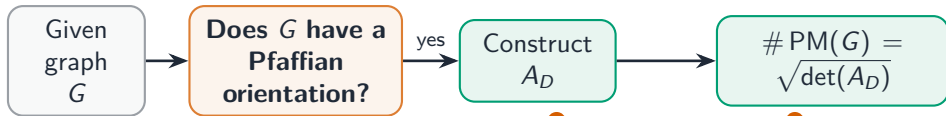
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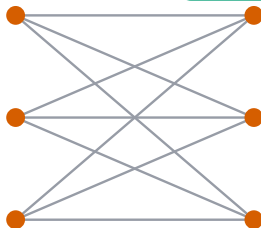
So far

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??

For example,



$K_{3,3}$

No polynomial-time algorithm
or hardness proof known

A route for every graph

What if one Pfaffian is not enough?

The **perfect-matching polynomial** of a graph G is defined as

$$\text{PM}(G) = \sum_{M \in \mathcal{M}(G)} \prod_{e \in M} x_e.$$

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But if G has no such orientations, one can combine Pfaffians from different orientations

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A graph admitting such a representation is **k -Pfaffian**. But how large can k be for graphs in general? The minimum number of orientations needed, that is k , is called the **Pfaffian number**. That is,

$$\text{pf}(G) = \min\{k : G \text{ is } k\text{-Pfaffian}\}.$$

A route for every graph

From planar graphs to higher genus

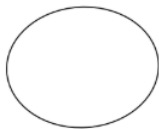
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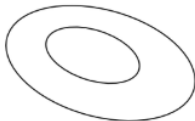
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$g = 0$



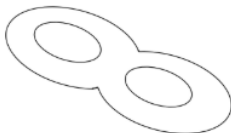
sphere

$g = 1$



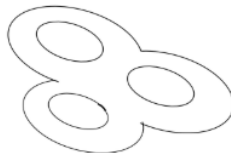
torus

$g = 2$



double torus

$g = 3$



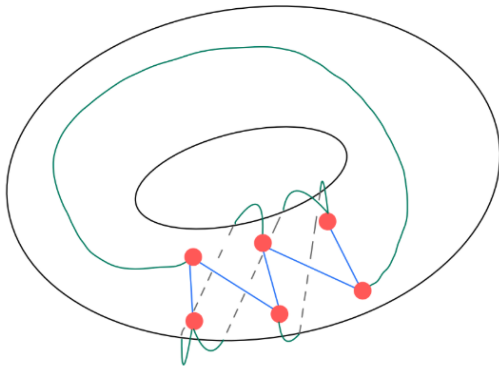
triple torus

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For example, $K_{3,3}$ has genus 1.



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Theorem (Galluccio–Loebl; Tesler)

Every graph of orientable genus g is 4^g -Pfaffian.

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Can the 4^g upper bound be improved to a subexponential bound?

A route for every graph

From planar graphs to higher genus

A graph G has orientable genus $\gamma(G) = g$ if G can be drawn without crossings on an orientable surface with g handles.

Theorem (Galluccio–Loebl; Tesler)

Every graph of orientable genus g is 4^g -Pfaffian.

Thus, one can count perfect matchings using at most 4^g determinants:

$$\text{pf}(G) \leq 4^g.$$

But what is the minimum number of pfaffians/determinants one needs?

Can the 4^g upper bound be improved to a subexponential bound?

Our result: No.

Our result

Exponential lower bounds for the Pfaffian number

Theorem

For every $g \geq 1$, there is a graph G with $\gamma(G) \leq g$ such that

$$\text{pf}(G) \geq \left\lceil \left(\frac{8}{3}\right)^g \right\rceil.$$

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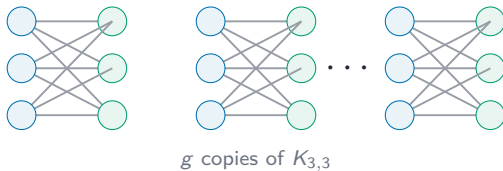
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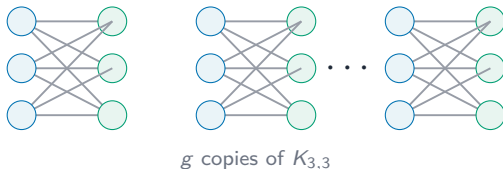
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And, any graph containing the above graph as a spanning subgraph inherits the lower bound.

How did we prove it?

The four polynomials

Hard polynomials

Matrix permanent

$$\text{per}(A) = \sum_{\pi \in S_n} \prod_{i=1}^n a_{i, \pi(i)}.$$

Perfect-matching polynomial

$$\text{PM}(G) = \sum_{M \in \mathcal{M}(G)} \prod_{e \in M} x_e.$$

Easy polynomials

Matrix determinant

$$\det(A) = \sum_{\pi \in S_n} \text{sgn}(\pi) \prod_{i=1}^n a_{i, \pi(i)}.$$

Pfaffian of an orientation D

$$\text{Pf}(A_D) = \sum_{M \in \mathcal{M}(G)} \text{sgn}_D(M) \prod_{e \in M} x_e.$$

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The four polynomials. For Bipartite graphs: PM = per

Let $G = (U \sqcup V, E)$, with $U = \{u_1, \dots, u_n\}$ and $V = \{v_1, \dots, v_n\}$. Define the biadjacency matrix $B_G = (b_{ij})$ by

$$b_{ij} = \begin{cases} x_{u_i v_j}, & u_i v_j \in E, \\ 0, & u_i v_j \notin E. \end{cases}$$

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Let D be an orientation of G . Its sign matrix is $S_D = (s_{ij}) \in \{\pm 1\}^{n \times n}$, where s_{ij} records the orientation of the edge $u_i v_j$.

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$$\text{sgn}_D(M_\pi) = \epsilon_n \text{sgn}(\pi) \prod_{i=1}^n s_{i, \pi(i)}, \quad \epsilon_n \in \{\pm 1\}.$$

*Note ϵ_n depends only on n , not on π .

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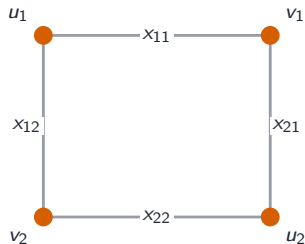
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$$\text{Pf}(A_D) = \epsilon_n \det(S_D \circ B_G).$$

A small example

Perfect matchings and the permanent

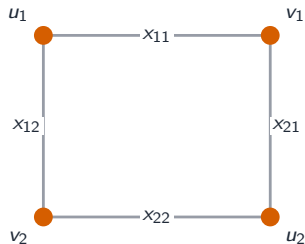
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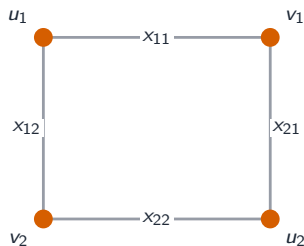
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$$\mathcal{M}(G) = \left\{ M_1 = \{u_1 v_1, u_2 v_2\}, M_2 = \{u_1 v_2, u_2 v_1\} \right\}.$$

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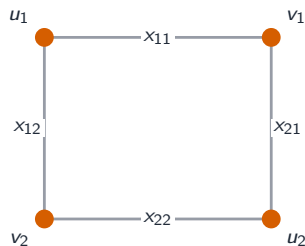
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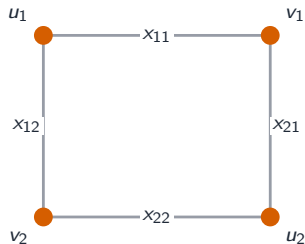
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The determinant and the Pfaffian

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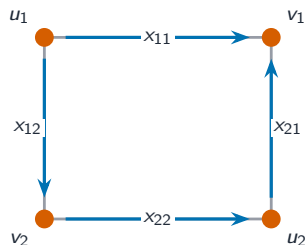


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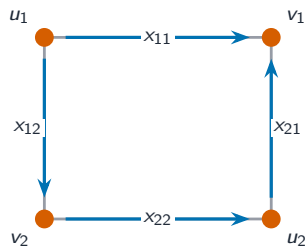
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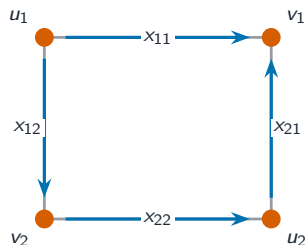
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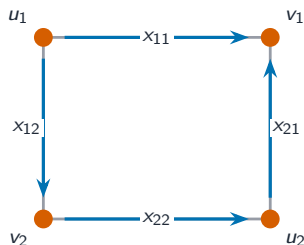
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Hence,

$$\text{PM}(G) = \text{per}(B_G) = \pm \text{Pf}(A_D) = -\det(S_D \circ B_G).$$

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Thus it suffices to show:

Theorem

Let A be an $n \times n$ matrix of distinct variables. If, as a polynomial identity,

$$\text{per}(A) = \sum_{i=1}^k c_i \det(S_i \circ A) \text{ then } k \geq \left\lceil \left(\frac{8}{3}\right)^{\lfloor n/3 \rfloor} \right\rceil.$$

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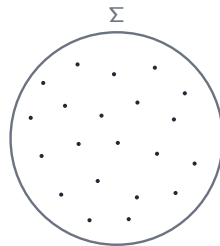
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Consider the family $\Sigma = \{\pm 1\}^{3 \times 3}$, $|\Sigma| = 512$.



512 sign matrices

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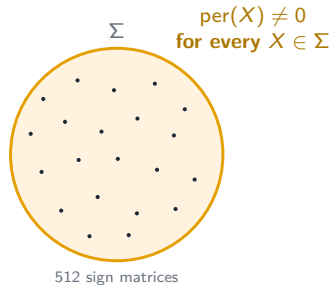
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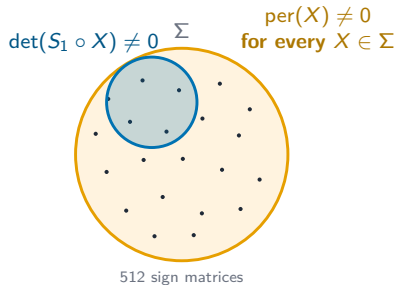
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Fact 2. For every fixed sign matrix $S \in \{\pm 1\}^{3 \times 3}$, $|\{X \in \Sigma : \det(S \circ X) \neq 0\}| = 192$.



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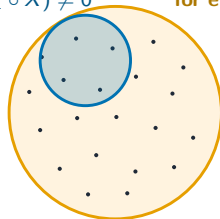
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Assume there exists an expression

$$\text{per}(A) = \det(S_1 \circ A) + \det(S_2 \circ A) + \cdots + \det(S_k \circ A).$$

$\det(S_1 \circ X) \neq 0$ Σ $\text{per}(X) \neq 0$
for every $X \in \Sigma$



512 sign matrices

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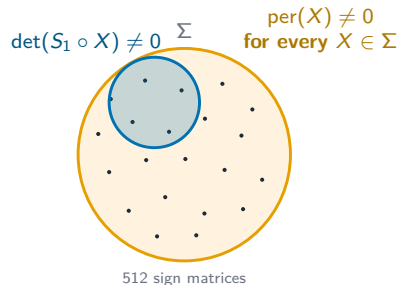
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LHS $\neq 0$ for all 512 matrices $A \in \Sigma$.



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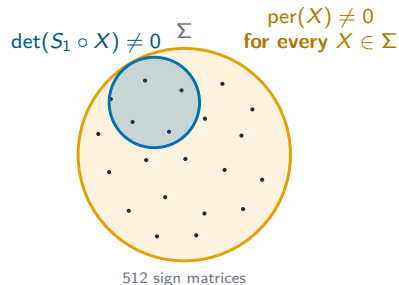
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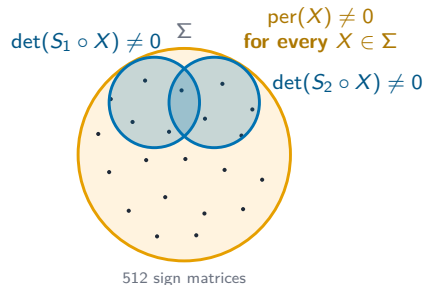
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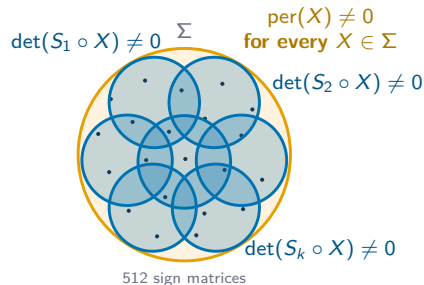
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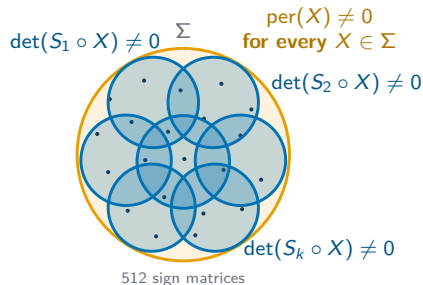
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For a given S_i , $\det(S_i \circ A) \neq 0$ for 192 out of 512 choices $A \in \Sigma$. Since there are k such S_i , thus

$$192k \geq 512 \implies k \geq \frac{8}{3}.$$



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The proof

Theorem

Let A be an $n \times n$ matrix of distinct variables. If, as a polynomial identity, $\text{per}(A) = \sum_{i=1}^k c_i \det(S_i \circ A)$ then $k \geq \left\lceil \left(\frac{8}{3}\right)^{\lfloor n/3 \rfloor} \right\rceil$.

Proof.

Now we amplify. Assume

$$\text{per}(A) = \det(S_1 \circ A) + \det(S_2 \circ A) + \cdots + \det(S_k \circ A)$$

Test it on $X \in \Sigma^g$.

LHS $\neq 0$ for all 512^g matrices $A \in \Sigma^g$.

But, $\det(S_1 \circ A) \neq 0$ for 192^g out of 512^g choices $A \in \Sigma^g$. Thus,

$$192^g k \geq 512^g \implies k \geq \left(\frac{8}{3}\right)^g. \quad \square$$

Let

$$X = \begin{pmatrix} X_1 & 0 & \cdots & 0 \\ 0 & X_2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & X_g \end{pmatrix}.$$

where each $X_j \in \Sigma$.

How did we prove it?

Back to graphs

The weighted biadjacency matrix of

$$G_g = \underbrace{K_{3,3} \sqcup K_{3,3} \sqcup \cdots \sqcup K_{3,3}}_{g \text{ copies}}.$$

has the same block-diagonal structure $B_1 \oplus \cdots \oplus B_g$; the X_i 's are the $\{\pm 1\}$ -evaluations used above.

Hence our matrix lower bound gives immediately $\text{pf}(G_g) \geq \left(\frac{8}{3}\right)^g$.

We also show:

Lemma

If H is a spanning subgraph of G , then $\text{pf}(G) \geq \text{pf}(H)$.

Since G_r is a spanning subgraph of $K_{3r,3r}$, $\text{pf}(K_{3r,3r}) \geq \text{pf}(G_r) \geq \left(\frac{8}{3}\right)^r$.

Previous: $\text{pf}(K_{3r,3r}) \geq 3r + 1$, **New:** $\text{pf}(K_{3r,3r}) \geq \left(\frac{8}{3}\right)^r$.

Open directions

- ▶ What other consequences follow from the fact that representing $\text{per}(A)$ by signed determinants requires exponentially many terms?
- ▶ What is the true exponential base?

$$\frac{8}{3} \leq \limsup_{g \rightarrow \infty} \text{pf}_{\max}(g)^{1/g} \leq 4.$$

- ▶ What is the correct growth rate for $\text{pf}(K_{n,n})$.

Thank you. Questions?